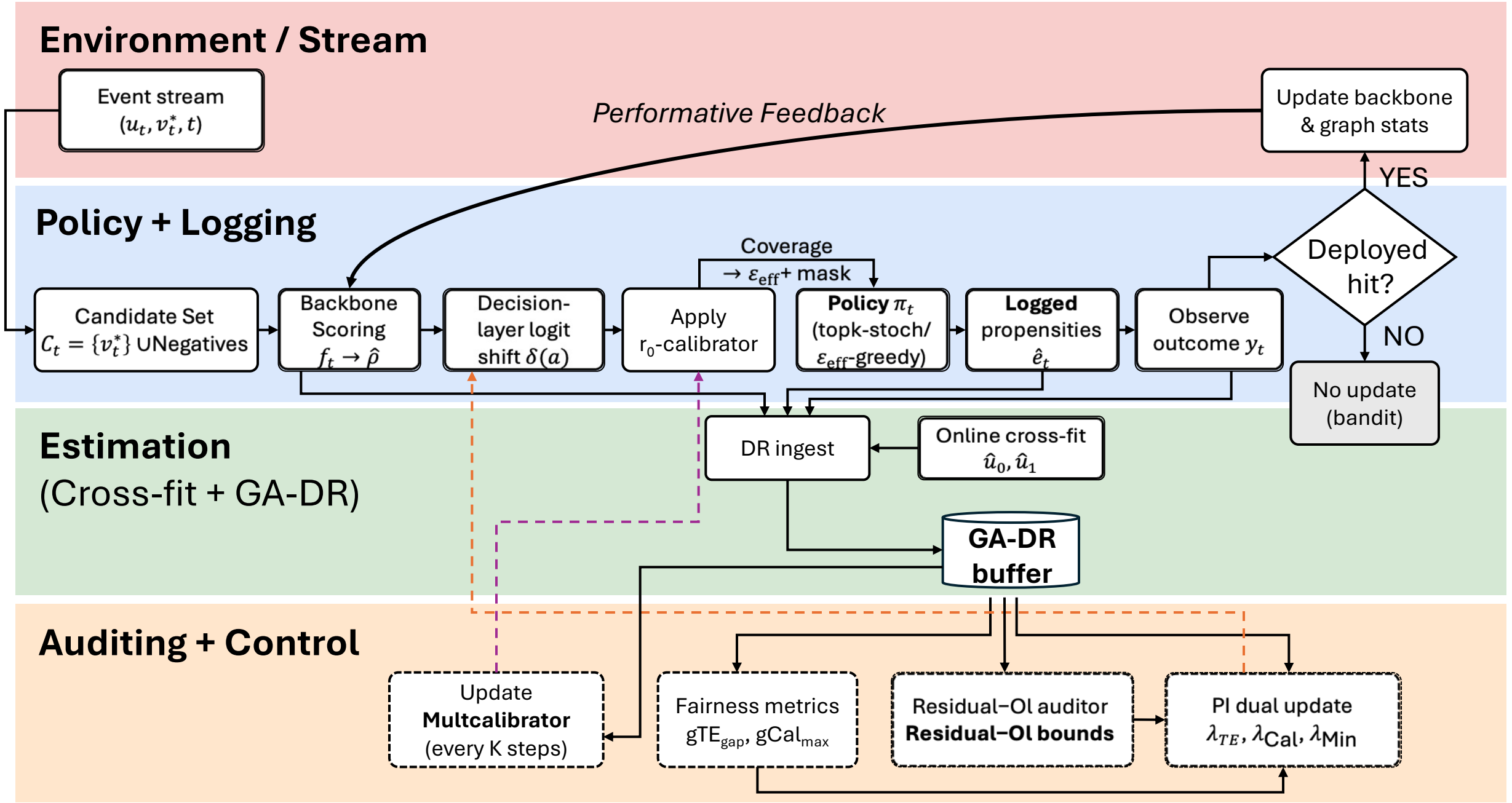

Environment / Stream
Event stream
$(u_t, v_t^*, t)$
Performative Feedback
Update backbone & graph stats
YES
Policy + Logging
Coverage
→ $\varepsilon_{\text{eff}}$+ mask
Deployed hit?
NO
Candidate Set
$C_t = \{v_t^*\}$ UNegatives
Backbone Scoring
$f_t \to \hat{\rho}$
Decision-layer logit shift $\delta(a)$
Apply $r_0$-calibrator
Policy $\pi_t$ (topk-stoch/ $\varepsilon_{\text{eff}}$-greedy)
Logged propensities $\hat{e}_t$
Observe outcome $y_t$
No update (bandit)
Estimation
(Cross-fit + GA-DR)
DR ingest
Online cross-fit
$\hat{u}_0, \hat{u}_1$
GA-DR buffer
Auditing + Control
Update Multcalibrator (every K steps)
Fairness metrics
$gTE_{gap}$, $gCal_{max}$
Residual–OI auditor
Residual–OI bounds
PI dual update
$\lambda_{TE}, \lambda_{Cal}, \lambda_{Min}$